Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/09/2025
PI Name:	Weeks, Eric R Ph.D.		
Project Title:	Thermal Fluctuations of Colloidal Gels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	erweeks@emory.edu	**Fax:**	FY 404-727-0873
PI Organization Type:	UNIVERSITY	**Phone:**	404-727-4479
Organization Name:	Emory University		
PI Address 1:	Physics Department		
PI Address 2:	400 Dowman Drive, Mail Stop 1131/002/1AB		
PI Web Page:	http://www.physics.emory.edu/		
City:	Atlanta	**State:**	GA
Zip Code:	30322-2430	**Congressional District:**	5
Comments:			
Project Type:	Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2020 Physical Sciences NNH20ZDA014N: Use of the NASA Physical Sciences Informatics System – Appendix G
Start Date:	12/01/2021	**End Date:**	11/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2024 per NSSC information (Ed., 10/7/24).		
Key Personnel Changes/Previous PI:	n/a		
COI Name (Institution):			
Grant/Contract No.:	80NSSC22K0292		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We studied colloidal gels using the data from the Advanced Colloids Experiment-Microscopy-1 (ACE-M-1) experiments. Colloidal gels are formed from sticky micron-sized solid particles in a liquid, where over time the particles stick together and form random clumps that eventually bridge across the entire sample chamber in tangled networks. The ACE-M-1 data set consists of a large number of optical microscopy movies of colloidal gels. In microgravity, these gels are long-lived, rather than collapsing under their own weight as happens in ground-based experiments. We have analyzed these microscopy movies using particle tracking, studying the visible thermal fluctuations and the gel structure as it evolves over time. Some gels have thick aggregated regions, while others have more tenuous gel strands and free particles. We show that the structure and dynamics change over the several-day period of each experiment. We have additionally conducted new ground-based microscopy studies of highly polydisperse colloidal gels, to understand how their structure differs from colloidal gels composed of similar-sized particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Colloids are suspensions of small solid particles suspended in a liquid. Examples include paint, ink, pastes, and blood. "Small" means the particle diameters range from ~10 nm to ~10 microns. Thermal motion is relevant: Brownian motion allows particles to diffuse. Often precautions are taken to prevent the solid particles from sticking together. If particles have attractive interactions, they can stick together in free-floating aggregates, or large tendrils that can span across the system. The latter is a colloidal gel. Colloidal gels are used in applications such as water purification, skin creams, and also show up in some food products such as jellies and jams. In food, colloidal gels modify the texture and shelf-life stability of the food. Our NASA-funded study of colloidal gels should improve our understanding of long-term stability of colloidal gels, as well as how they initially form. One project we are pursuing is the study of colloidal gels made from highly polydisperse particles: particles of a wide range of sizes. This is more realistic for industrial materials, as compared to a lot of prior work which studied particles that were are nearly the same size. Thus, our work will help understand the structure of these more realistic, easier to make colloidal gels.
Task Progress:	We have submitted two manuscripts for publication. Both preprints are publicly available: • “Microstructure of polydisperse colloidal gels,” Benjamin F. Lonial and Eric R. Weeks, preprint (arXiv: 2406.10321) -- we submitted this to Phys. Rev. E and are currently revising it in response to referee comments. • “Aging of colloidal gels in microgravity,” Swagata S. Datta, Waad Paliwal, and Eric R. Weeks, preprint (arXiv: 2501.09650) -- we submitted this to Phys. Rev. E, we revised it in response to referee comments, and it is currently under review again at that journal. [Ed. Note: See Cumulative Bibliography for citations.] Our project primarily involves analyzing data from NASA’s Physical Systems Informatics (PSI) database. In particular, we analyzed microscope images of colloidal gels from the ACE-M-1 experiment (Advanced Colloids Experiment). The ACE-M-1 experiment resulted in 530 GB of images, which we successfully downloaded in 2022. Colloids are small (micron-sized) particles in a liquid. In these experiments, the particles are made to be sticky, so they stick together into a network of tendrils of particles. This is the colloidal gel state. The ACE-M-1 experiment was done on the International Space Station so that the colloidal gels could be studied in microgravity conditions. The reason microgravity is important is that, as the particles begin to stick together, they form heavy clusters which can sink in normal gravity. This ultimately limits the shelf-life of a colloidal gel, but also, changes the overall structure. We wish to understand what the gel structure would be like if this limit is overcome in microgravity conditions. RESULTS FOR MICROGRAVITY DATA: Under such microgravity conditions, we observe a slowdown in particle dynamics consistent with gel aging. Stronger attractive forces promote the formation of thicker gel strands over time. The ACE-M1 samples are bidisperse, composed of particles with a size ratio 1 : 1.2. Larger particles experience stronger depletion forces, which lead to a prevalence of large-large particle contacts. As the gel ages, small mobile particles are incorporated into the gel structure. The changes in gel structure correlate with a slow decay in particle motion, observed over time scales ranging from tens of minutes to tens of hours. Additionally, through complementary ground-based experiments, we compare two-dimensional (2D) and three-dimensional (3D) images of depletion colloidal gels. While microgravity gel data are limited to 2D projections, ground-based data establish a correspondence between the 2D and 3D data on particle contacts. Our results provide new insights into how colloidal gels age in the absence of gravitational collapse. NEW GROUND-BASED EXPERIMENTS ON HIGHLY POLYDISPERSE COLLOIDAL GELS: Ben Lonial and Eric Weeks made colloidal gels with particles with extreme size polydispersity. Our goal is to understand how the mixture of particle sizes changes the gel structure. While the ACE-M-1 data uses two different particle sizes, they are fairly similar in size (1.8 micron and 2.2 micron diameter) and the gel structure does not vary a huge amount due to these sizes. On the other hand, with the larger range of sizes in Ben’s new samples, we see differences. The largest particles are more than 15 times the size of the smallest particles. The volume fractions studied ranged from 0.01 to 0.50; at the latter, half of the volume of the sample is solid particles and the other half the liquid solvent. The pairing of nearest neighbor particles is consistent with a null hypothesis that pairings are made randomly; that is, any two particle sizes have a probability of being neighbors, consistent with how many of each we have. That being said, as expected, larger particles have more nearest neighbors than small ones because they have more surface area to stick to neighbors. This leads to an over-representation of large particles in tetrahedral structures where four particles are mutually nearest neighbors, showing that large particles help provide rigidity to the gel structure. Our preprint discusses the implications of how other mixtures of sizes would affect the gel structure.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Other Journals or Periodicals	Datta SS, Paliwal W, Weeks ER. "Aging of colloidal gels in microgravity." arXiv preprint server. Posted January 16, 2025. https://doi.org/10.48550/arXiv.2501.09650 , Jan-2025

Articles in Peer-reviewed Journals	Datta SS, Paliwal W, Weeks ER. "Aging of colloidal gels in microgravity." Phys. Rev. E. 2025 Jul;112(1):015412. https://doi.org/10.1103/vdgy-2f6p , Jul-2025
Articles in Peer-reviewed Journals	Lonial BF, Weeks ER. "Microstructure of polydisperse colloidal gels." Physical Review E. 2026 Feb;113(2):025420. https://doi.org/10.1103/kmpn-zjk7 , Feb-2026